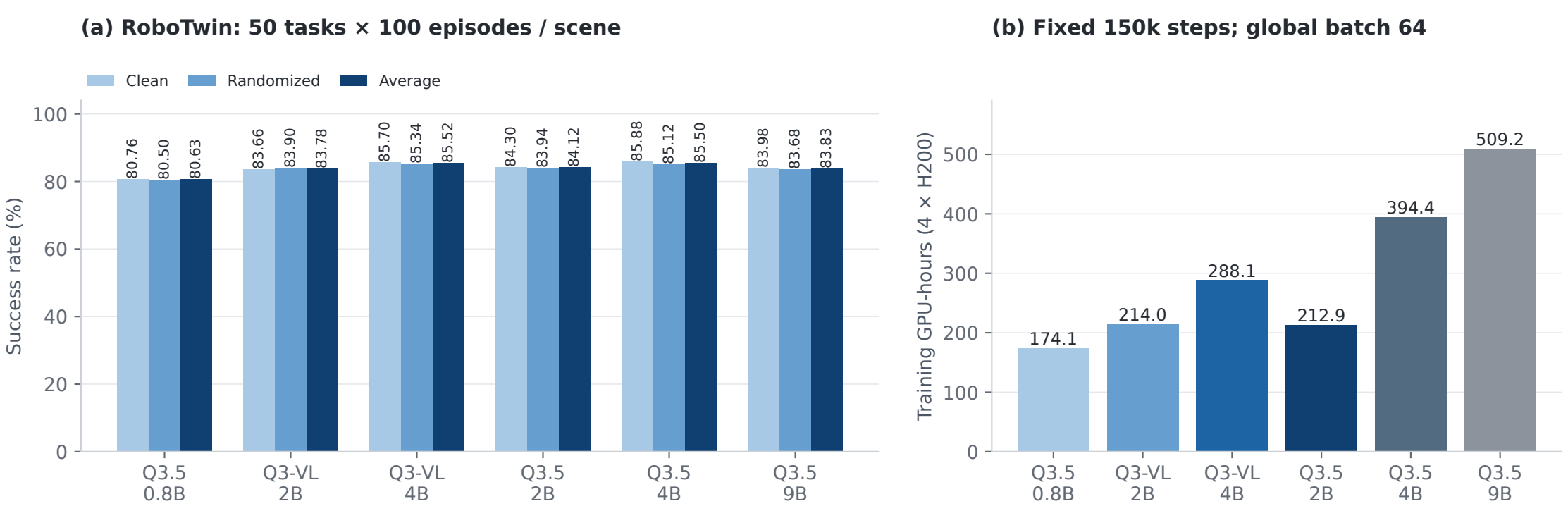
(a) RoboTwin: 50 tasks × 100 episodes / scene
Clean
Randomized
Average
Success rate (%)
0
20
40
60
80
100
80.76
80.50
80.63
83.66
83.90
83.78
85.70
85.34
85.52
84.30
83.94
84.12
85.88
85.12
85.50
83.98
83.68
83.83
Q3.5 0.8B
Q3-VL 2B
Q3-VL 4B
Q3.5 2B
Q3.5 4B
Q3.5 9B
(b) Fixed 150k steps; global batch 64
Training GPU-hours (4 × H200)
0
100
200
300
400
500
174.1
214.0
288.1
212.9
394.4
509.2
Q3.5 0.8B
Q3-VL 2B
Q3-VL 4B
Q3.5 2B
Q3.5 4B
Q3.5 9B